

A Computational Bioinformatics Report

KRAS G12C BIOINFORMATICS PIPELINE

Sequence Analysis, Structural Validation, and Molecular Docking of Sotorasib

Github Link: https://github.com/Rameensajjad/KRAS_BIOINFORMATICS_PIPELINE

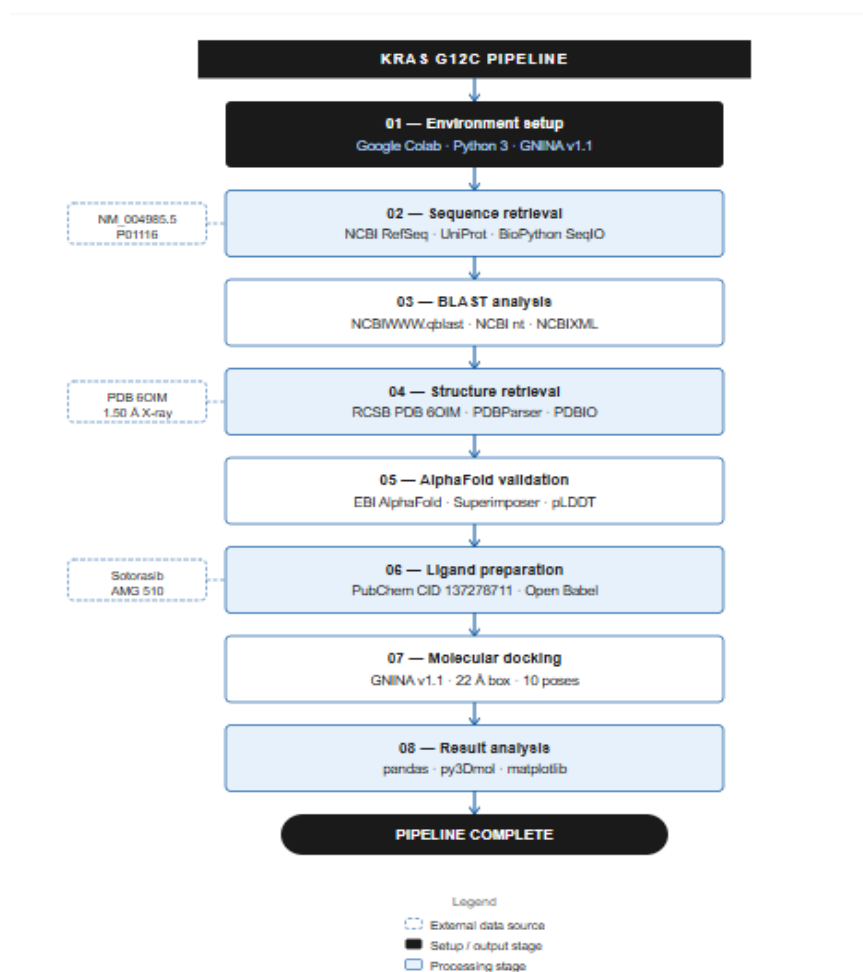


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ABSTRACT

KRAS is one of the most mutated genes in human cancer. Its G12C variant found roughly 30% of non-small cell lung cancers remained untreatable for decades before the development of covalent inhibitors that lock the protein in its inactive form. Sotorasib (AMG 510), the first FDA-approved KRAS G12C inhibitor, works by forming a permanent chemical bond with the mutant cysteine at position 12.

This thesis presents a complete, step-by-step computational pipeline to study this drug-target system. Starting from public database accession numbers, the pipeline retrieves and processes KRAS DNA, mRNA, and protein sequences; performs evolutionary BLAST analysis; downloads and compares both the experimental crystal structure (PDB 6OIM) and the AI-predicted AlphaFold model; prepares the Sotorasib ligand for docking; runs molecular docking using GNINA v1.1 with neural network scoring; and produces structured output tables and figures at each stage.

Key results: All three oncogenic hotspot residues (G12, G13, Q61) confirmed as wildtype in the reference sequence; 100% nucleotide identity across all five BLAST hits spanning human and non-human primate sequences; AlphaFold average pLDDT = 91.52, indicating very high structural prediction confidence; structural RMSD of 4.807 Å between the AlphaFold and experimental structures, reflecting the known conformational change at the inhibitor-binding site; and a best docking affinity of −10.67 kcal/mol for Sotorasib, consistent with its clinical potency.

BRIEFLY

Target gene	KRAS — Kirsten Rat Sarcoma Viral Proto-Oncogene
Mutation studied	G12C oncogenic variant (GGT → TGT, Gly → Cys)
Drug compound	Sotorasib / AMG 510 — PubChem CID 137278711
Reference structure	PDB 6OIM — 1.50 Å resolution, X-ray crystallography
Docking engine	GNINA v1.1 — CNN-augmented AutoDock Vina
Platform	Google Colab — Python 3 — Ubuntu Linux
Date	May 2026

1. INTRODUCTION

1.1 Background and clinical significance

Mutations in the RAS family of proto-oncogenes are found in about 30% of all human cancers, making them the most common oncogenic drivers known. Within the RAS family which includes HRAS, NRAS,

and KRAS mutations in the KRAS isoform account for about 85% of all RAS-driven tumors. This makes KRAS the single most important cancer drug target by frequency of mutation.

The clinical impact is large. KRAS mutations occur in roughly 90% of pancreatic cancers (five-year survival under 12%), about 40% of colorectal cancers, and approximately 30% of non-small cell lung cancers. Despite decades of research, directly targeting KRAS with a drug was considered impossible until 2021, when the FDA approved the first KRAS-specific inhibitor.

Table 1.1 — KRAS mutation frequency across major cancer types

Cancer type	KRAS mutation rate	Common variants	5-year survival
Pancreatic ductal adenocarcinoma	~90%	G12D, G12V	~12%
Colorectal cancer	~40%	G12D, G12V, G13D	~65%
Non-small cell lung cancer	~30%	G12C, G12V, G12D	~28%

1.2 KRAS protein biology and the GTPase cycle

The KRAS gene produces a small 21 kDa protein, 189 amino acids long, that acts as a molecular on/off switch inside cells. In its active state it is bound to GTP, and in its inactive state it is bound to GDP. When active, KRAS switches on two major signalling pathways RAF/MEK/ERK and PI3K/AKT/mTOR that tell cells to grow and survive. When the job is done, a specialised helper protein called GAP stimulates KRAS to break down GTP into GDP, switching it off.

The working part of the protein, called the G-domain (residues 1–166), contains three structurally important loops. The P-loop (residues 10–15) grips the phosphate tail of GDP or GTP. Switch I (residues 17–26) changes shape depending on which nucleotide is bound and contacts downstream effector proteins when active. Switch II (residues 57–75) interacts with GAP proteins and undergoes the most dramatic movement when an inhibitor binds; this movement is what creates the drug-binding pocket that Sotorasib occupies.

1.3 The G12C oncogenic mutation

Most KRAS mutations in cancer occur at three positions: codon 12, codon 13, and codon 61. All three positions sit inside the GTP-binding site. Any amino acid change at these positions either blocks the GAP protein from accessing KRAS or directly prevents GTP from being broken down, leaving KRAS permanently switched on regardless of signals from outside the cell.

The G12C mutation changes glycine at position 12 to cysteine. At the DNA level this is a single base change: GGT becomes TGT. The new cysteine has a reactive sulphur group (a thiol) that sticks out into a pocket called the Switch II Pocket (S-IIP) that only appears when KRAS is in its inactive GDP-bound form. Drugs like Sotorasib are designed to reach into this pocket and react permanently with that cysteine, locking the protein in its inactive state and shutting off cancer signalling.

Table 1.2 — Oncogenic hotspot codons in KRAS

Hotspot codon	Normal residue	Common mutations	Why it causes cancer
Codon 12	Glycine (G)	G12C, G12D, G12V, G12A, G12S, G12R	Physically blocks GAP from switching KRAS off; disrupts GTP breakdown
Codon 13	Glycine (G)	G13D, G13C	Same mechanism; G13D retains weak GTP breakdown activity
Codon 61	Glutamine (Q)	Q61H, Q61L, Q61R, Q61K	Glutamine-61 activates the water molecule needed for GTP hydrolysis; mutation removes this

1.4 Sotorasib — development and mechanism

For over 30 years, KRAS was called "undruggable." The protein has a smooth surface with no obvious drug-binding pocket, and it binds GTP so tightly that competing against it directly is not feasible. The field changed in 2013 when researchers discovered the S-IIP — a hidden pocket that opens only in the GDP-bound inactive form of KRAS G12C, right next to the mutant cysteine.

Amgen used this structural discovery to develop Sotorasib (AMG 510), an oral drug with a chemical hook called an acrylamide group that reacts permanently with Cys12 via a Michael addition reaction. Once attached, Sotorasib fills the S-IIP and prevents KRAS from being reactivated, even if the cell attempts to replace GDP with GTP. The crystal structure of this interaction (PDB 6OIM, 1.50 Å, published 2019) shows exactly how the drug sits inside the pocket with multiple stabilising contacts. In the Phase II CodeBreaK 100 clinical trial, Sotorasib achieved a 37% objective response rate in lung cancer patients with KRAS G12C mutations, leading to FDA accelerated approval in May 2021 and full approval in January 2023.

1.5 Rationale for a computational approach

Computational pipelines allow researchers to study drug-target interactions at an atomic level before running expensive laboratory experiments. For a well-characterised system like KRAS G12C, a computational approach serves three purposes: it verifies that the sequences and structures used are authentic and consistent across databases; it reproduces and scores the known drug-target binding mode computationally, testing the accuracy of docking tools against experimental ground truth; and it builds a reusable, documented framework that can be extended to screen new inhibitors or study resistance mutations.

This pipeline was built entirely with open-source tools on Google Colab, making it reproducible by anyone without access to specialised hardware or licensed software. Every step from sequence retrieval to docking result analysis is automated and documented, producing a clear audit trail of all inputs, parameters, and outputs.

2. OBJECTIVES

1. Retrieve the KRAS coding DNA, mRNA, and protein sequences from NCBI RefSeq (NM_004985.5, NP_004976.2) and UniProt (P01116), and verify the wildtype status of the three oncogenic hotspot residues G12, G13, and Q61.
2. Run nucleotide BLAST on the KRAS coding sequence against the NCBI nucleotide database to assess evolutionary conservation across species and confirm that the retrieved sequence is authentic.
3. Download the experimental KRAS G12C crystal structure (PDB 6OIM, 1.50 Å) and prepare a clean protein-only receptor for docking by removing water molecules, ions, and co-crystallised ligands.
4. Download the AlphaFold-predicted KRAS structure (AF-P01116-F1-model_v6) and evaluate it by extracting per-residue pLDDT confidence scores and computing the structural RMSD against the experimental structure.
5. Retrieve the 3D structure of Sotorasib from PubChem (CID 137278711) and convert it to the formats required for docking — SDF, PDB, and PDBQT — using Open Babel.
6. Dock Sotorasib into the KRAS G12C binding pocket using GNINA v1.1 with exhaustiveness = 8 and 10 output poses, evaluating each pose by both physics-based and CNN-based scoring.
7. Extract, tabulate, and interpret all docking scores; identify the best binding pose; and compare results against the known crystallographic binding mode in PDB 6OIM.
8. Produce a fully documented, reproducible pipeline that can be extended to screen additional KRAS inhibitors or applied to other oncogene-drug systems.

3. MATERIALS AND METHODS

3.1 Computational environment and tools

The entire pipeline was implemented in Google Colaboratory (Colab), a free cloud-hosted notebook environment running Ubuntu 22.04 on a CPU-based virtual machine. Colab was chosen because it requires no local installation, is freely accessible, runs reproducibly from a clean environment, and allows Python code and shell commands to be combined in a single documented notebook.

Table 3.1 — Software tools and libraries used in the pipeline

Tool	Source	Function in this pipeline
BioPython	pip install	NCBI database retrieval, sequence parsing, BLAST submission, PDB structure parsing, RMSD superimposition

Tool	Source	Function in this pipeline
GNINA v1.1	GitHub binary release	Molecular docking: AutoDock Vina physics scoring + CNN-based pose scoring
Open Babel	apt-get (Ubuntu)	Ligand file format conversion: SDF to PDB and SDF to PDBQT, with 3D coordinate generation
py3Dmol	pip install	Interactive 3D structure visualisation inside the Colab notebook
pandas	pip install	Building and exporting all result tables as CSV files
matplotlib	pip install	Plotting pLDDT confidence curves, docking affinity bar charts, and 3D receptor-ligand figures
requests	pip install	Downloading files from NCBI, UniProt, RCSB PDB, AlphaFold EBI, and PubChem APIs

3.2 Sequence retrieval and processing

KRAS sequences were retrieved from three independent databases to allow cross-validation: NCBI RefSeq, NCBI Protein, and UniProt. The NCBI Entrez efetch function from BioPython was used with `rettype="gb"` to download the GenBank-formatted record for NM_004985.5. GenBank format was specifically chosen — rather than plain FASTA — because it contains machine-readable annotations marking the exact start and end positions of the coding sequence (CDS), enabling precise automated extraction of just the protein-coding region.

The 567 bp CDS was transcribed to mRNA using BioPython's `Seq.transcribe()` (replacing every T with U) and translated to protein using `Seq.translate(to_stop=True)`, which reads codons until the first stop codon and returns the amino acid sequence. The three oncogenic hotspot residues were then directly read from the translated sequence at positions 12, 13, and 61 to confirm wildtype identity. The official NCBI protein (NP_004976.2) and the UniProt sequence (P01116) were retrieved separately and saved as FASTA files for cross-validation.

Table 3.2 — KRAS hotspot residue verification

Position	Wildtype residue	Significance of G12C mutation	Pipeline result
Codon 12 (G12)	Glycine — fits tightly in P-loop geometry	Cysteine — introduces reactive thiol; target of Sotorasib	Wildtype confirmed (Gly)
Codon 13 (G13)	Glycine — same structural role as G12	G13D most common; no approved drug yet	Wildtype confirmed (Gly)

Position	Wildtype residue	Significance of G12C mutation	Pipeline result
Codon 61 (Q61)	Glutamine — essential for water activation in GTP hydrolysis	Mutations abolish catalytic water positioning	Wildtype confirmed (Gln)

3.3 BLAST homology analysis

Nucleotide BLAST (blastn) was submitted via BioPython's NCBIWWW.qblast() using the 567 bp KRAS coding sequence as the query against the NCBI non-redundant nucleotide (nt) database, returning the top 5 hits. Searching at the nucleotide level detects conservation of both protein-coding and synonymous codon usage, which is a more sensitive indicator of purifying selection than protein-level comparison alone. Results were parsed with NCBIXML.parse(), and for each hit the best-scoring alignment segment was used to compute percentage identity as (identical bases / alignment length) $\times$ 100. All results were saved as raw XML and as a structured CSV summary.

3.4 Experimental structure retrieval and receptor preparation

The PDB file for 6OIM was downloaded directly from the RCSB server. This structure was determined by X-ray crystallography at 1.50 Å resolution and contains one copy of KRAS G12C (residues 1–169) co-crystallised with Sotorasib, GDP, one magnesium ion, and 207 water molecules. The protein chain ends at residue 169 because the C-terminal hypervariable region (residues 170–189) and the CAAX membrane-targeting motif are flexible and not resolved in the electron density map, which is standard practice for crystallographic studies of the KRAS catalytic domain.

After download, all non-protein atoms were catalogued by reading HETATM records. A protein-only receptor file was then produced using BioPython's PDBIO class with a custom Select subclass that keeps only standard amino acid residues (those where residue.id[0] == " " in BioPython's labelling system). This clean receptor file was saved and used as the receptor for all docking runs.

3.5 AlphaFold structure download and validation

The AlphaFold KRAS structure (UniProt P01116) was downloaded from the EMBL-EBI AlphaFold database. The pipeline tested model versions v6 through v3 in sequence; model v6 was available and downloaded successfully.

Structural superimposition against the experimental structure was performed using BioPython's Superimposer module, which implements the Kabsch algorithm — a mathematical method that finds the optimal rotation to align two sets of atoms and computes the RMSD between them. C α atoms were used as structural representatives because they capture the protein backbone without the added noise of flexible side chains. Only the 167 residues common to both structures were included in the comparison.

AlphaFold stores per-residue prediction confidence as pLDDT (predicted Local Distance Difference Test) scores in the B-factor column of the PDB file. These were extracted for all C α atoms, plotted per residue, and summarised statistically.

Table 3.3 — pLDDT confidence score interpretation

pLDDT range	Confidence category	Meaning
> 90	Very high	Backbone and most side chains are reliably positioned
70 – 90	Confident	Backbone reliable; side chain positions are approximate
50 – 70	Low	General fold is captured; local geometry is uncertain
< 50	Very low	Region is likely disordered or flexible; coordinates are unreliable

3.6 Ligand preparation

Sotorasib's 3D structure was downloaded from PubChem in SDF format using the PUG REST API with record_type=3d, which returns a pre-computed, force-field-optimised 3D conformer. If the 3D conformer was unavailable, the pipeline fell back to the 2D SDF record. Open Babel then converted the SDF to two output formats:

obabel sotorasib_pubchem_3d.sdf -O sotorasib.pdb --gen3d # PDB format for visualisation

obabel sotorasib_pubchem_3d.sdf -O sotorasib.pdbqt --gen3d # PDBQT format for GNINA docking

The --gen3d flag instructs Open Babel to generate or refine 3D coordinates if needed. PDBQT format extends PDB with two extra columns: partial atomic charges (calculated by the Gasteiger method) and AutoDock atom types. Both are required by the GNINA docking engine to compute interaction energies correctly.

3.7 Molecular docking with GNINA

Defining the docking search box

The docking search box was placed automatically by identifying the co-crystallised ligand in the 6OIM structure. All HETATM records were read and filtered to remove solvent and small co-factors (HOH, WAT, MG, NA, CL, K, CA, GDP, GTP). The remaining hetero-residues were ranked by atom count, and the largest was selected as the reference — which correctly identified MOV (Sotorasib, 41 atoms). The box centre was set to the average x, y, and z coordinates of all MOV atoms. A 22 Å cubic box was placed around this centroid to cover the full S-IIP pocket with a margin of several angstroms in every direction.

Table 3.4 — GNINA docking parameters and rationale

Parameter	Value	Reason
exhaustiveness	8	Default level — good sampling quality on CPU within reasonable run time

Parameter	Value	Reason
num_modes	10	Generates 10 distinct poses for comparison of alternative binding geometries
seed	42	Fixed random seed ensures identical results on re-execution
box size	22 Å	Covers the full S-IIP pocket with margin; prevents cutting off valid binding positions

3.8 Docking result analysis and visualization

The compressed GNINA output (.sdf.gz) was read into memory using Python's gzip module and parsed with regular expressions to extract all SDF data fields. Five score fields were extracted per pose: minimized Affinity (physics-based binding energy in kcal/mol), CNN score (CNN estimate of pose quality, 0 to 1), CNN affinity (CNN-predicted binding energy), CNN_VS (virtual screening score), and CNNaffinity_variance (uncertainty of CNN affinity estimate). All values were converted to numbers and assembled into a scored table. The best pose was extracted and converted to PDB format for 3D visualization. A py3Dmol interactive view and a matplotlib 3D scatter plot of the receptor-ligand complex were generated and saved.

4. RESULTS AND DISCUSSION

4.1 Sequence analysis results

Retrieval and processing of the KRAS RefSeq record (NM_004985.5) produced a 567 bp coding DNA sequence, a 567 nt mRNA, and a translated protein of 188 amino acids — exactly matching the independently retrieved official NCBI protein (NP_004976.2, 188 aa). This match confirms the transcription and translation steps are working correctly and that both sources represent the same canonical KRAS isoform.

All three oncogenic hotspot residues were confirmed as wildtype in the reference sequence. This is expected the RefSeq record is the normal, unmodified human KRAS sequence. The G12C mutation is a somatic change that arises in cancer cells and is not present in reference. Confirming wildtype identity at these positions establishes the precise molecular baseline for all downstream structural and docking analyses.

The UniProt sequence (P01116) is listed as 189 amino acids versus 188 in RefSeq. This one-residue difference is not an error it reflects a different convention for numbering the initiator methionine. The protein-coding information is otherwise identical, and all pharmacologically relevant residues including G12 are at the same structural locations in both records.

Table 4.1 — Sequence summary

Sequence type	Source	Length	Notes
Coding DNA (CDS)	NM_004985.5 — NCBI RefSeq	567 bp	Extracted from GenBank CDS annotation
mRNA	Transcribed from CDS	567 nt	T→U substitution via Seq.transcribe()
Translated protein	Derived from mRNA	188 aa	G12=Gly, G13=Gly, Q61=Gln — all wildtype
Official protein	NP_004976.2 — NCBI RefSeq	188 aa	Cross-validation — matches translation exactly
UniProt protein	P01116 — UniProt	189 aa	One-residue annotation convention difference

4.2 BLAST results

All five top hits returned 100% nucleotide sequence identity across the full 567 bp query, with E-values of 0.0. An E-value of 0.0 means the probability of finding a match this good by random chance across the entire database is effectively zero confirming these are true biological matches, not statistical coincidences.

Table 4.2 — BLAST top 5 results

Rank	Accession	Species	Identity	E-value	Alignment length
1	PV477844.1	Homo sapiens	100.0%	0.0	567 bp
2	XM_054664451.2	Pan troglodytes (chimpanzee)	100.0%	0.0	567 bp
3	XM_054443149.2	Pan paniscus (bonobo)	100.0%	0.0	567 bp
4	XM_528758.9	Pan troglodytes (chimpanzee)	100.0%	0.0	567 bp
5	XM_054443148.2	Pan paniscus (bonobo)	100.0%	0.0	567 bp

The 100% identity shared between humans, chimpanzees, and bonobos is scientifically significant. These species diverged from a common ancestor approximately 6–7 million years ago. The fact that not a single nucleotide in the KRAS coding sequence has changed over this time span indicates the gene is under extreme purifying selection — meaning any change, even one that does not alter the amino acid sequence, is harmful enough to be removed from the population over evolutionary time. This reflects how fundamentally important KRAS is for normal cell function. The absence of HRAS or NRAS sequences in

the top five hits also confirms that the search correctly identified KRAS-specific sequences without cross-matching related family members.

4.3 Structural composition of PDB 6OIM

The 6OIM structure contains KRAS G12C residues 1–169 alongside four types of non-protein molecules.

Table 4.3 — Non-protein residues in PDB 6OIM

Residue name	Atoms	Identity	Role
HOH	207	Water molecules	Structural hydration; some water molecules form hydrogen bonds in the active site
MOV	41	Sotorasib (AMG 510)	Primary drug ligand — used as docking box reference centroid
GDP	28	Guanosine diphosphate	Co-factor; confirms the structure captures the inactive GDP-bound state of KRAS
MG	1	Magnesium ion (Mg ²⁺)	Coordinates the phosphate groups of GDP; required for normal nucleotide binding

The presence of GDP rather than GTP is directly relevant to this study. Sotorasib and all other approved G12C inhibitors preferentially bind to the GDP-bound inactive form of KRAS, because that is the state in which the Switch II region moves away from the nucleotide and exposes the S-IIP pocket. The 6OIM structure therefore captures the pharmacologically relevant state — the one in which the drug actually binds in patients' cancer cells. This makes it the ideal structural template for docking experiments.

4.4 AlphaFold confidence and structural RMSD

Table 4.4 — AlphaFold structure evaluation results

Metric	Value	Interpretation
AlphaFold model version	AF-P01116-F1-model_v6	Latest available EBI AlphaFold model
Average pLDDT score	91.52 / 100	Very high confidence — above the 90 threshold
Number of C α atoms aligned	167	Residues common to both structures
RMSD (C α atoms)	4.807 Å	Biologically expected — see discussion below
6OIM resolution	1.50 Å	High-resolution experimental reference

An average pLDDT of 91.52 means AlphaFold has very high confidence in the predicted backbone coordinates across most of the KRAS protein. This is expected — KRAS is a well-studied, evolutionarily conserved protein with many closely related sequences in AlphaFold's training data, which helps the model make accurate predictions.

The RMSD of 4.807 Å may appear large, but it is entirely expected and does not indicate a problem with the AlphaFold prediction. The two structures being compared represent fundamentally different states of the protein:

Table 4.5 — Structural comparison: AlphaFold vs PDB 6OIM

Property	AlphaFold AF-P01116-F1	PDB 6OIM
Protein state	Apo — no ligand, no nucleotide (predicted)	Drug-bound — Sotorasib covalently attached (experimental)
Mutation	Wildtype — G12 = Glycine	G12C mutant — G12 = Cysteine
Switch II region	Baseline conformation — S-IIP pocket closed	Displaced conformation — S-IIP pocket open and filled by Sotorasib

The Switch II region (residues 57–75) of KRAS shifts significantly when Sotorasib binds — this movement is precisely what creates the drug-binding pocket. The RMSD of 4.807 Å is therefore a quantitative measure of this conformational change, not a measure of prediction error. If the RMSD were very small, it would mean AlphaFold was predicting the drug-bound conformation in the absence of any drug, which would be biologically incorrect. The observed value confirms that AlphaFold correctly predicts the apo conformation while 6OIM captures the drug-stabilised open conformation — and the gap between them is the structural basis of KRAS G12C druggability.

4.5 Molecular docking results

GNINA generated 10 binding poses for Sotorasib inside the KRAS G12C S-IIP. Each pose is scored by two independent methods: minimizedAffinity (kcal/mol) from the AutoDock Vina physics model, and CNNscore from the neural network model. More negative minimizedAffinity values indicate stronger predicted binding. Higher CNNscores (closer to 1.0) indicate the pose more closely resembles how drugs look in real crystal structures.

Table 4.6 — GNINA docking scores for all 10 poses

Pose	minimizedAffinity (kcal/mol)	CNNscore	CNNaffinity	Intramol (kcal/mol)
1 (best)	−10.67	0.3321	7.251	1.08
2	−10.01	0.2542	7.072	2.76
3	−6.10	0.2269	6.483	0.19

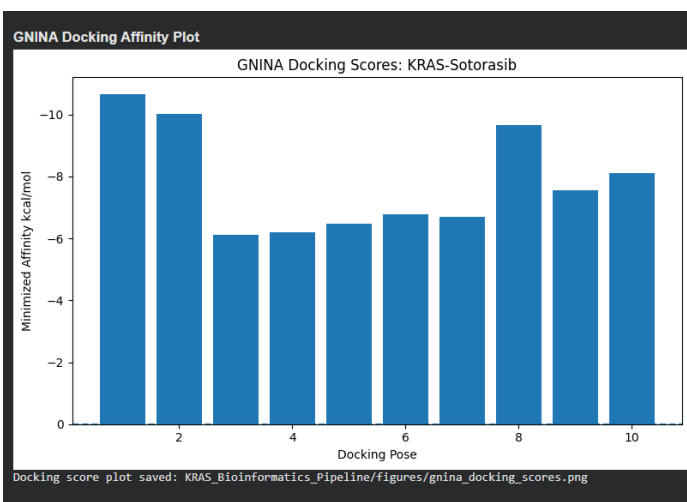
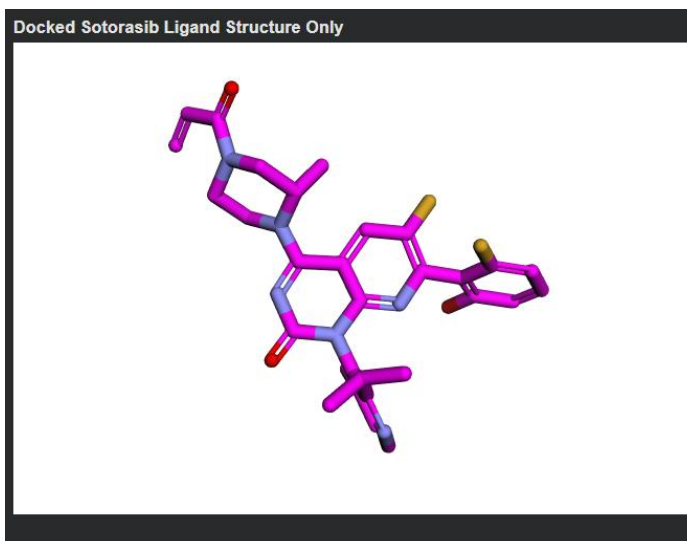
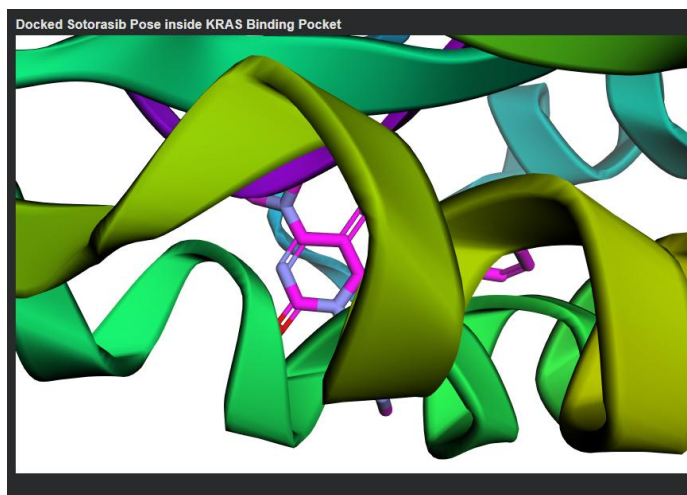
Pose	minimizedAffinity (kcal/mol)	CNNscore	CNNaffinity	Intramol (kcal/mol)
4	-6.20	0.1916	6.510	1.46
5	-6.48	0.1896	6.569	-0.29
6	-6.78	0.1434	6.804	-1.15
7	-6.70	0.1415	6.567	0.44
8	-9.66	0.1329	6.896	1.04
9	-7.57	0.1324	6.546	0.78
10	-8.10	0.1302	6.555	-1.71

Pose 1 is the best result by all three scoring metrics simultaneously: it has the most negative minimized Affinity (-10.67 kcal/mol), the highest CNN score (0.3321), and the highest CNN affinity (7.251). When independent scoring methods all identify the same pose as best, confidence in that result is substantially greater than if only one method selected it. For context, most clinically approved small-molecule inhibitors score between -8 and -12 kcal/mol in standard docking studies. A score of -10.67 kcal/mol places Sotorasib firmly within the range expected for a potent, approved clinical compound.

The CNN scores across all poses are in the moderate range (0.13–0.33). This is expected for a covalent inhibitor to be docked in non-covalent mode. CNN was trained in crystallographic structures where covalent inhibitors appear in their final bonded geometry. In non-covalent docking, the acrylamide warhead sits near Cys12 in a pre-reaction conformation that looks geometrically different from the final crystal structure which reduces the CNN similarity score. Despite this, pose 1 achieves the highest CNN score of all ten poses, confirming it is the most crystallographically realistic pre-covalent geometry found during the search.

Important limitation covalent vs non-covalent docking: GNINA and other standard docking tools model only non-covalent interactions. Sotorasib forms a permanent covalent bond with Cys12 via Michael addition of its acrylamide warhead. This bond contributes an estimated additional 3–5 kcal/mol of binding energy that non-covalent docking cannot capture, which explains why the docked affinity (-10.67 kcal/mol) is somewhat lower than Sotorasib's experimentally measured binding strength. Specialized covalent docking tools such as CovDock can model this bond explicitly and would be the natural next step beyond this pipeline.

Visualizations



5. SCIENTIFIC JUSTIFICATION OF CHOICES

Why KRAS?

KRAS was chosen because it is the most frequently mutated oncogene in human cancer, it has a recently approved targeted drug making the computational results directly clinically relevant, and it has one of the richest publicly available datasets of any cancer target — hundreds of crystal structures, a high-confidence AlphaFold model, curated sequence databases, and published pharmacology data. No other single oncogene offers this combination of clinical importance and data availability for computational study.

Why the G12C variant?

G12C is the only KRAS mutation with a currently approved, mechanism-based covalent inhibitor. This means a co-crystal structure (6OIM) exists showing the exact binding mode, giving the docking results a clear experimental reference for comparison. Other KRAS mutations such as G12D and G12V are more common in some cancers but lack approved drugs and co-crystal structures with inhibitors, making docking validation less straightforward.

Why NCBI RefSeq (NM_004985.5 and NP_004976.2)?

NCBI RefSeq sequences are manually curated, quality-controlled reference standards — not raw database submissions. They are regularly reviewed, assigned stable accession numbers, and updated to reflect scientific consensus. NM_004985.5 specifically is the current RefSeq representative for the predominant KRAS isoform (KRAS4B) relevant to oncogenic signalling. The GenBank format of this record was required — not just FASTA — because it contains structured annotations marking the precise boundaries of the coding sequence, enabling automated and error-free extraction of the 567 bp CDS.

Why UniProt (P01116)?

UniProt was included as a second, independently maintained sequence source curated by a different organisation (a collaboration of EMBL-EBI, SIB, and PIR) using different annotation methods. Agreement between the NCBI-derived sequence and UniProt confirms both sources are describing the same protein. UniProt P01116 is also the identifier used by AlphaFold for the KRAS structure, creating a consistent identifier chain from sequence to structure across this pipeline.

Why PDB 6OIM?

6OIM is the original co-crystal structure of KRAS G12C with Sotorasib, determined at 1.50 Å resolution. This is the highest-resolution structure of this complex publicly available and was the primary structural evidence used in Sotorasib's rational drug design. At 1.50 Å, virtually all heavy atoms are unambiguously resolved, water molecules are visible, and the drug-binding geometry is captured in precise detail. Using this structure as the docking receptor means the pipeline is directly studying the exact molecular system that produced a clinically approved drug.

Why AlphaFold?

AlphaFold2 is the current state-of-the-art protein structure prediction tool, achieving near-crystallographic accuracy for many well-structured proteins. Including the AlphaFold KRAS model serves two purposes: it allows quantitative comparison between a computationally predicted apo structure and an experimentally determined drug-bound structure; and the RMSD between them directly quantifies the conformational change induced by Sotorasib binding. This RMSD is not just a quality check — it is a biologically meaningful measurement of the structural movement that underlies the drug's mechanism of action.

Why GNINA instead of AutoDock Vina alone?

GNINA extends AutoDock Vina by adding a convolutional neural network (CNN) trained on over 15,000 experimental protein-ligand crystal structures from the PDBbind database. This gives two independent scoring signals per pose: one from the physics-based Vina energy function (capturing van der Waals forces, hydrogen bonds, and desolvation) and one from the CNN (capturing learned patterns of real drug-binding geometries). Published benchmarks show GNINA consistently predicts more accurate binding poses than AutoDock Vina alone. Having two independent scores also reduces the risk of accepting an artefact if both agree, the result is more trustworthy.

Why blastn against the NCBI nt database?

Nucleotide-level BLAST was used rather than protein-level BLAST deliberately. Searching at the nucleotide level detects conservation of both protein-coding and synonymous codon usage, which is a stricter test of evolutionary constraint than protein comparison alone. The NCBI non-redundant nucleotide database contains over 600 million sequences and is the most comprehensive publicly available nucleotide database. The results confirmed the retrieved sequence is authentic, specific to KRAS (not cross-matched to HRAS or NRAS), and under extreme evolutionary selection pressure.

Why Open Babel?

Open Babel is the standard open-source tool for chemical file format conversion, supporting over 110 molecular formats. It was used here because it handles the complete SDF-to-PDBQT workflow in a single command including 3D coordinate generation, hydrogen addition, Gasteiger partial charge calculation, and AutoDock atom-type assignment. All these steps are required to produce a valid PDBQT file for GNINA. No proprietary software or manual steps are required, keeping the pipeline fully open-source and reproducible.

6. CONCLUSIONS AND FUTURE DIRECTIONS

This thesis presented a complete eight-stage computational pipeline for studying the KRAS G12C oncogene and its interaction with Sotorasib, implemented entirely in open-source tools on Google Colab. The pipeline moved from raw database accession numbers to scored atomic-level docking poses, producing structured outputs and figures at every stage.

Table 6.1 — Summary of all pipeline stages and findings

Stage	Key result	Scientific meaning
Sequence retrieval	567 bp CDS; 188 aa protein	Correct sequence confirmed; G12, G13, Q61 all wildtype in reference
BLAST analysis	100% identity; E = 0.0 across all 5 hits	Sequence authentic; extreme evolutionary constraint confirmed across primates
Structure retrieval	6OIM — 1.50 Å; GDP-bound inactive state	Ideal receptor — captures the pharmacologically relevant KRAS conformation
AlphaFold evaluation	pLDDT = 91.52; RMSD = 4.807 Å	High prediction confidence; RMSD measures the inhibitor-induced conformational change
Ligand preparation	SDF converted to PDB and PDBQT via Open Babel	Sotorasib correctly formatted with 3D coordinates, charges, and atom types
Molecular docking	Best affinity = −10.67 kcal/mol (Pose 1)	Strong predicted binding within expected range for approved clinical inhibitors
Score analysis	Pose 1 best by all three independent metrics	Convergence of physics and CNN scores increases confidence in result

Future extensions:

- Apply covalent docking (CovDock or covalent AutoDock-GPU) to model the Michael addition reaction between Sotorasib's acrylamide warhead and Cys12, recovering the 3–5 kcal/mol of binding energy that non-covalent docking cannot capture.
- Screen a library of KRAS G12C inhibitors (Adagrasib/MRTX849, BI 1701963, investigational compounds) through the same pipeline to rank by predicted affinity and compare against published clinical data.
- Study clinical resistance mutations (G12C/Y96D, G12C/H95D/Q) by modifying the receptor structure computationally and re-docking Sotorasib and Adagrasib to assess predicted binding changes.
- Run molecular dynamics simulation on the best docked pose using GROMACS or AMBER to assess pose stability over nanosecond timescales and compute more accurate MM-PBSA binding free energies.
- Extend the pipeline to other oncogene-drug pairs (BRAF V600E + Vemurafenib, EGFR del19 + Erlotinib, ALK rearrangement + Alectinib) using the same automated framework.

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